

Vulnerability of information transport on temporal networks to link removal

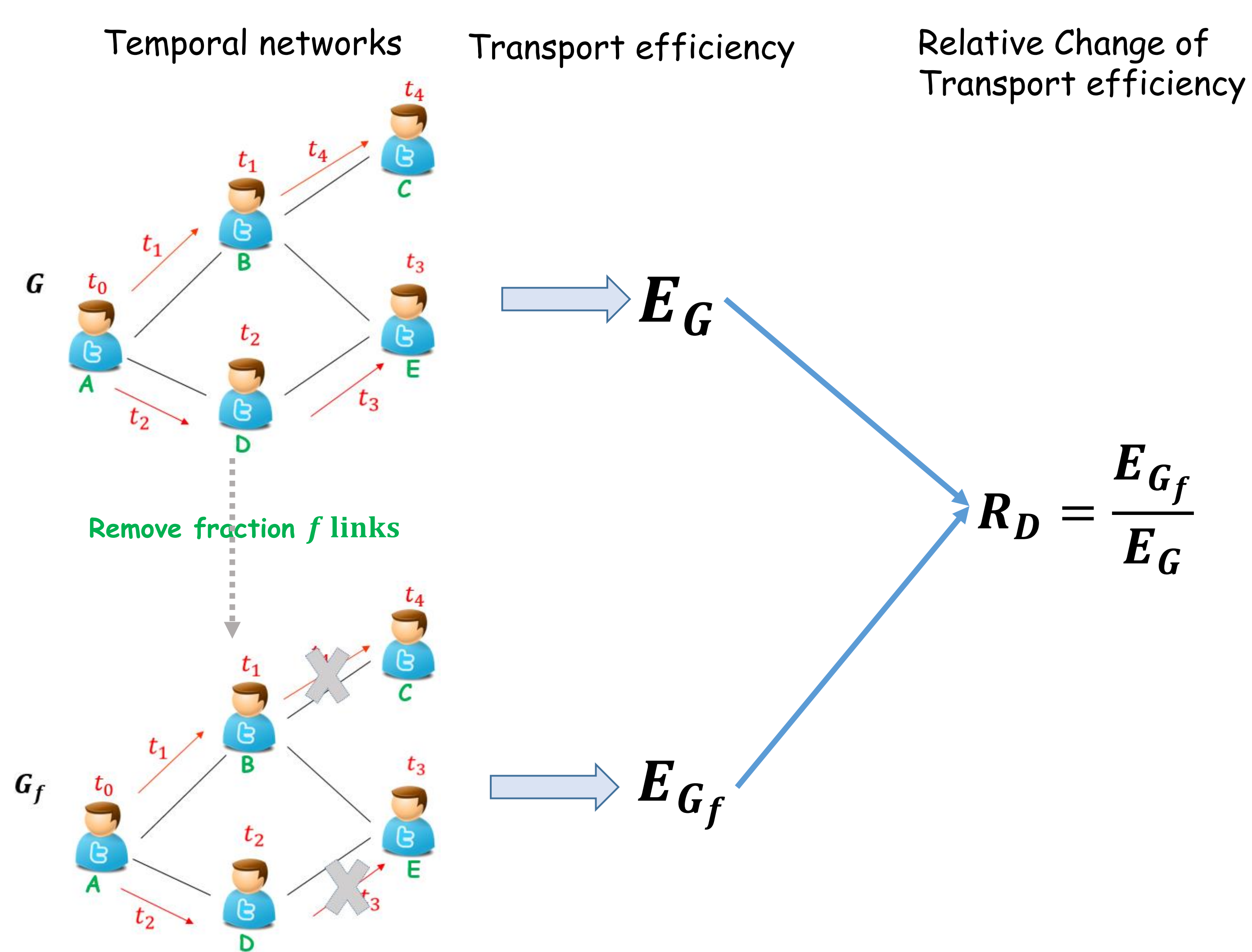
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Motivation & Research Goals

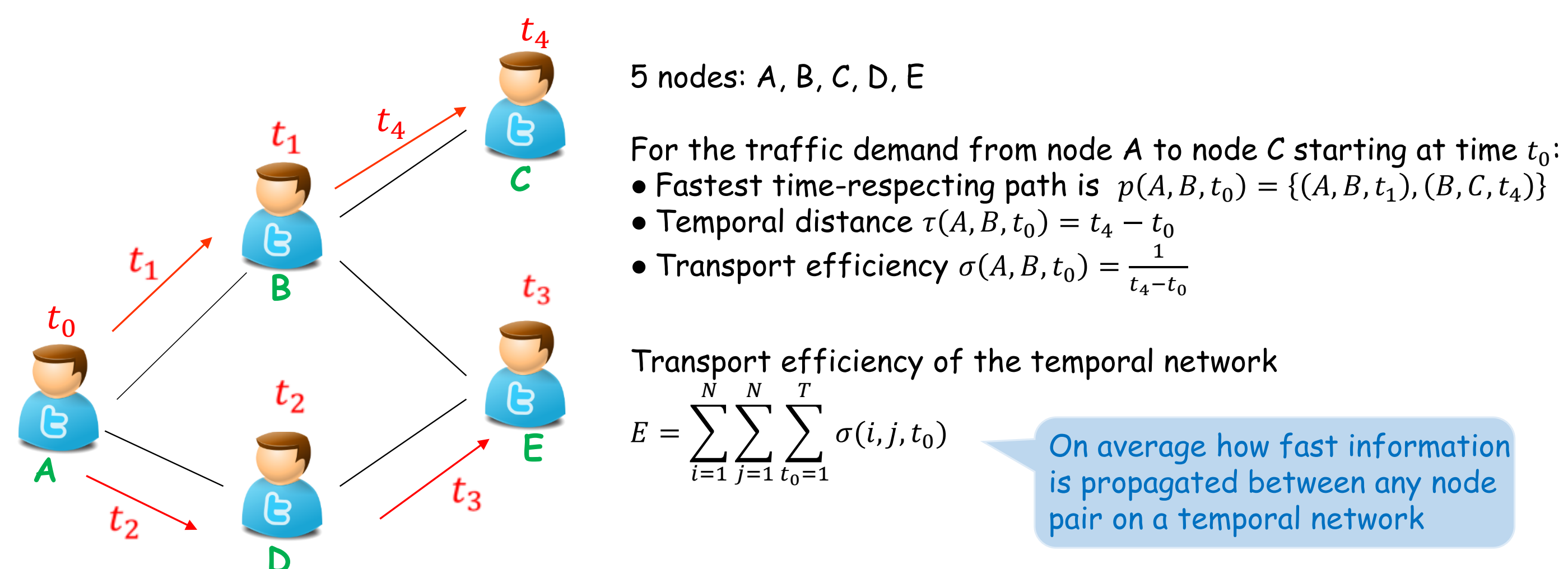
- Communications networks such as opportunistic mobile networks, vehicle networks, and social contact networks can be represented as temporal networks, where two mobile devices are connected only when they are close to each other. These networks facilitate the propagation of information.
- In this work, we investigate the vulnerability of information transport on a temporal network to link removal. The objective is to understand the removal of which types of links deteriorate the efficiency of information transport the most.
- Identifying such critical links via their properties will enable better protection (intervention) to facilitate (prohibit) the spread of (mis)information.

Methods

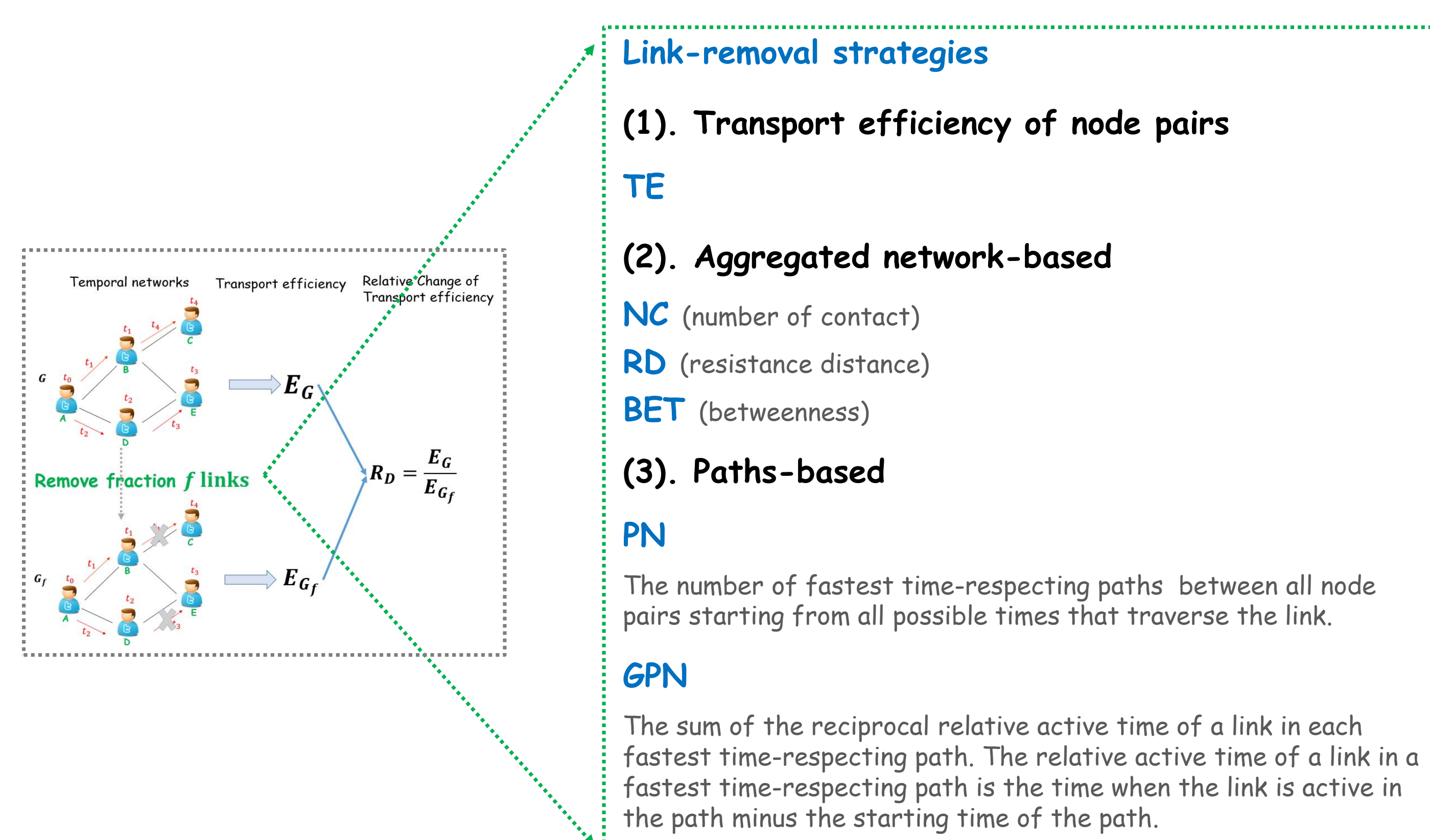
(1) Outline



(2) Transport efficiency of temporal networks

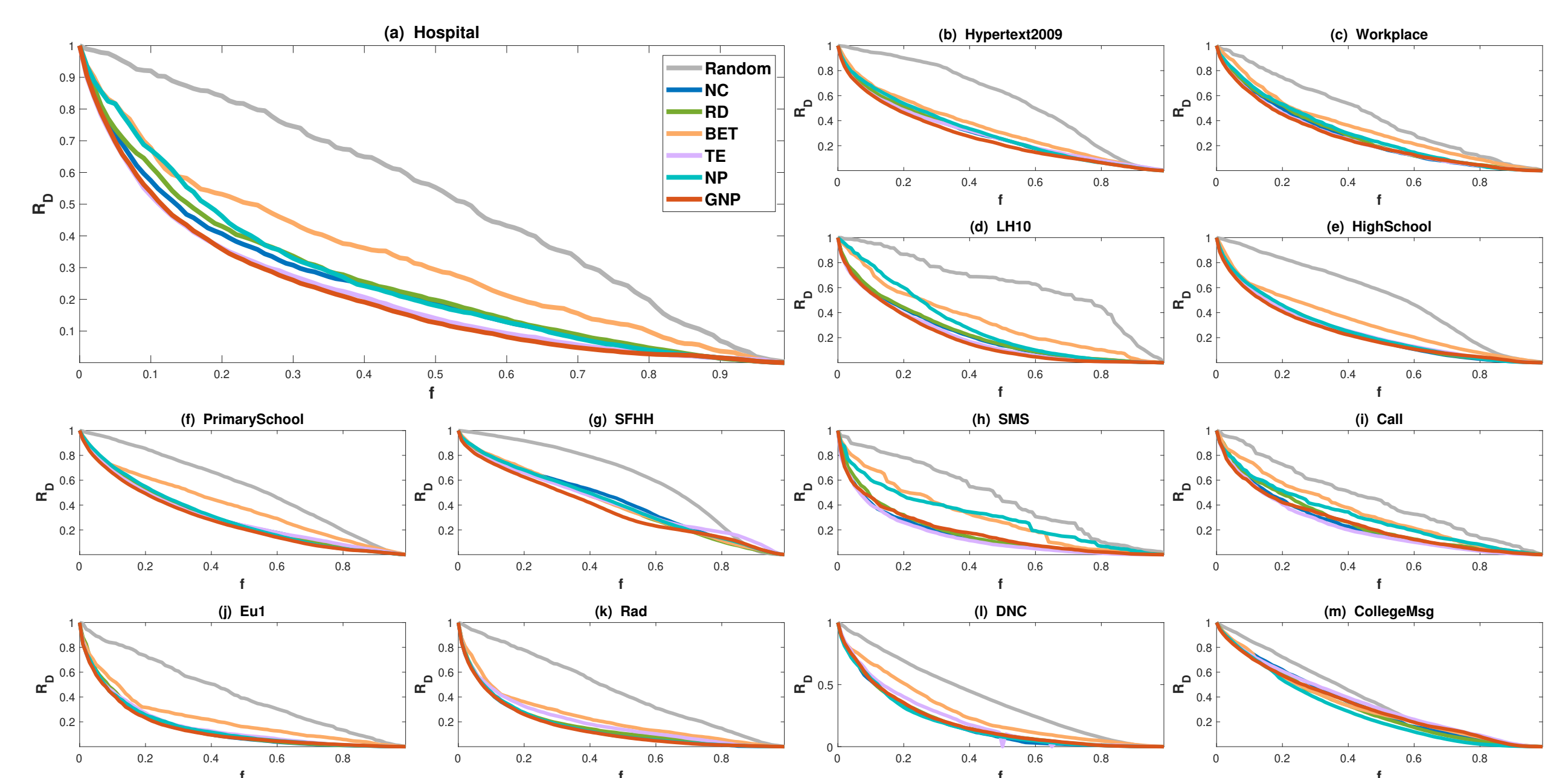


(3) Link-removal strategies



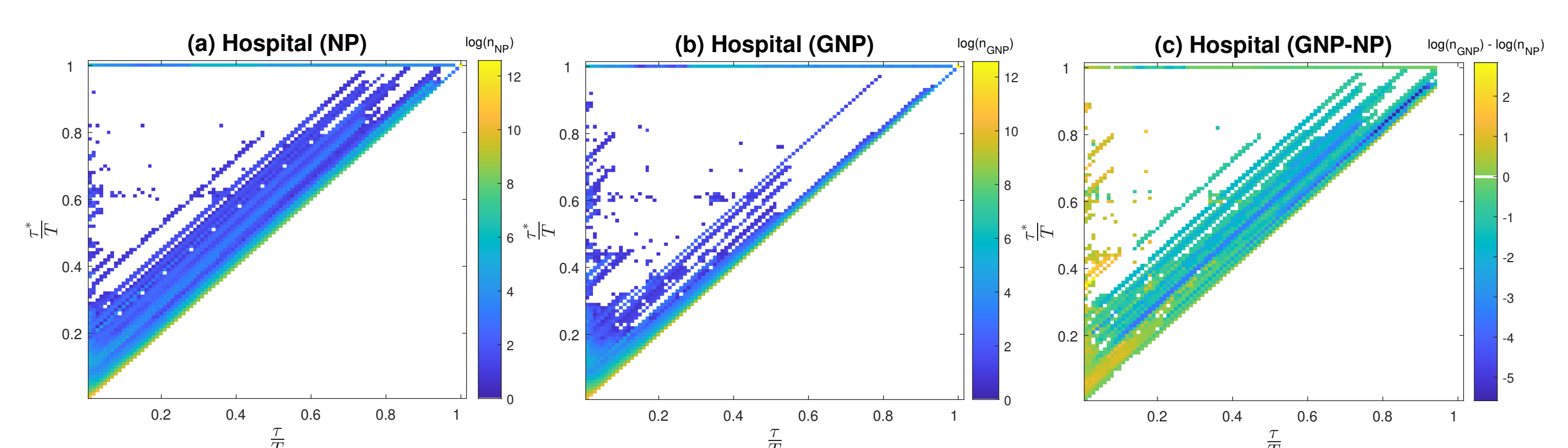
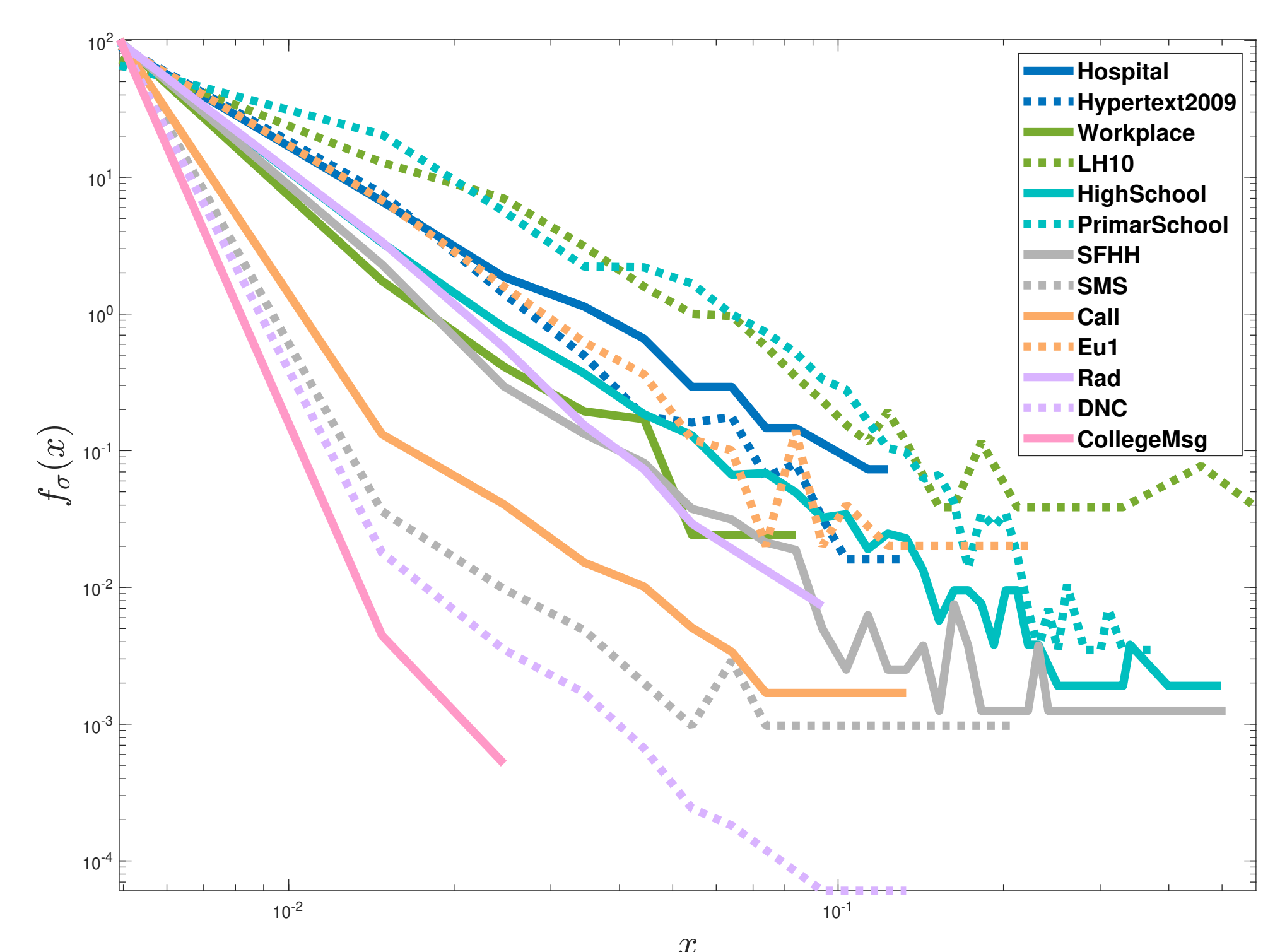
Selected Results

(4) Evaluation



This figure shows the relative change of transport efficiency $R_D(f)$ as a function of the fraction f of links removed according to different link-removal strategies. Our proposed strategies can reduce the transport efficiency of the network more significantly than random removal. The most effective strategy for identifying critical links is GNP.

(5) Explanation



- Why is the transport efficiency of physical contact networks higher than that of virtual contact networks?
- Why does GNP work better than NP?